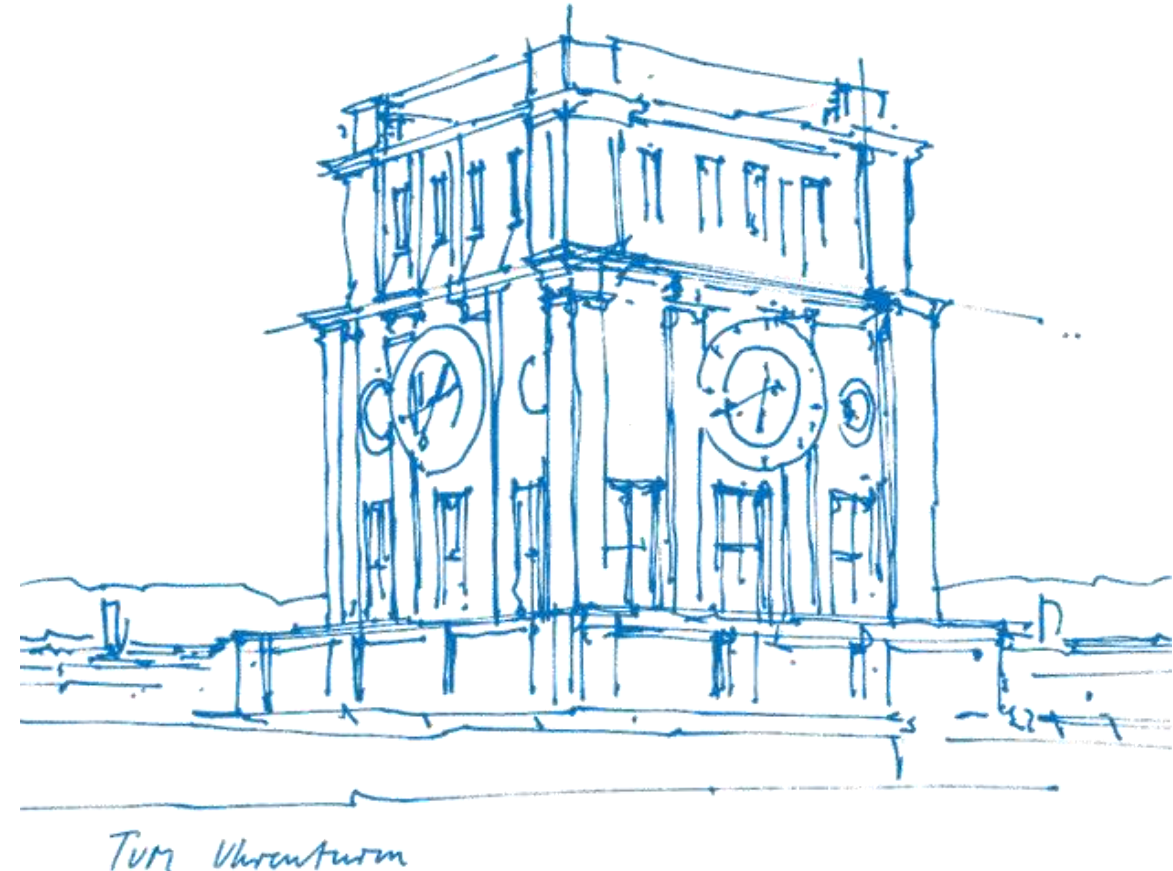


Tutorial 05

Memoryless Distributions, Joint and Marginal pdf

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About Tutorial

■ **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
- 14:15-15:45 Tue, 00.08.053 (B42)

■ The slides are not guaranteed to be accurate! (Written by myself)

■ **Materials of this tutorial:** Please scan the QR-Code 

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- **PART I.** Recap of the Lecture
- **PART II.** Exercises from the Exercise Sheet
- **PART III.** Additional Exercises (with solution)



I. Recap

- **Memoryless Distributions**

- Definition
- Discrete memoryless distribution (Geometric distribution)
- Continuous memoryless distribution (Exponential distribution)

- **Joint and Marginal pdf**

- Definition

II. Exercise Sheet

Exercise 1 (value at risk (VaR)). An investment analyst models the daily percentage return R of a particular stock as a continuous random variable with pdf

$$p_R(r) = \begin{cases} \frac{2}{15}(r+3) & \text{if } -3 \leq r < 0, \\ \frac{1}{5}(2-r) & \text{if } 0 \leq r \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

Since r is in percent, $r = 1$ means a 1% return. The *Value at Risk (VaR)* at confidence level α (or, expressed in percentages, at a $(100\alpha)\%$ confidence level) is the maximum percentage loss L such that the probability of experiencing a loss greater than L is α . Mathematically, if $L > 0$ is the VaR, then $P(R \leq -L) = \alpha$.

- (i) Verify that $p_R(r)$ is a valid pdf.
- (ii) Calculate the cumulative distribution function (CDF), $F_R(r)$. Be sure to define it for all $r \in \mathbb{R}$.
- (iii) Explain how the $(100\alpha)\%$ VaR (expressed as a positive loss percentage) relates to the α -quantile, $r_\alpha = F_R^{-1}(\alpha)$, of the return distribution R .
- (iv) Determine the 5% Value at Risk (VaR) for this investment.
- (v) Calculate the median daily return

II. Exercise Sheet

Exercise 2 (lack of memory of the exponential distribution). Let $X : (\Omega, \mathcal{A}, P) \rightarrow (\mathbb{R}_{>0}, \mathcal{B})$ be a random variable such that $X > 0$ with strictly positive density function $p_X(x) > 0$ for all $x \in \mathbb{R}_{>0}$. Show that X is exponentially distributed, i.e., $X \sim \text{Exp}(\lambda)$ for some $\lambda > 0$, if and only if

$$P(X > t + s \mid X > s) = P(X > t) \quad \forall s, t \geq 0. \quad (\text{A})$$

[Hint: Note that the cdf of $\text{Exp}(\lambda)$ is $F_X(x) = 1 - e^{-\lambda x}$ and that $P(X > x) = 1 - P(X \leq x) = 1 - F_X(x)$. Use that the conditional probability can be written as $P(X > t + s)/P(X > s)$. For “ $\Leftarrow$ ” consider $P(X > x) = \int_x^\infty f_X(a) da =: f(x)$ and $\varphi(s) = \frac{f(t+s)}{f(s)}$. How does φ depend on s ?]

II. Exercise Sheet

Exercise 3. Let X and Y be two jointly continuous random variables with joint pdf

$$p_{X,Y}(x, y) = \begin{cases} \frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6} & \text{if } x \in [0, 1], y \in [0, 2], \\ 0 & \text{otherwise.} \end{cases}$$

For $y \in [0, 2]$, find

- (i) the conditional pdf of X given $Y = y$, i.e., the function $p_{X|Y}(x | y)$,
- (ii) the probability $P(X < \frac{1}{2} | Y = y)$.

III. Additional Exercises

Exercise 4. In a test, the DPT tutor Angelika gives her students three tasks. For each correctly solved task, Angelika awards one point. If a task is solved incorrectly, no point is given for that task. Assume that the students earn points randomly in the test. Let X_i be the score from the i -th task, and let

$$f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} (x_1 + x_2 + x_3 - 1)^2 / 8 & \text{for } x_1, x_2, x_3 \in \{0, 1\} \\ 0 & \text{otherwise} \end{cases}$$

be the joint density.

Determine the marginal densities f_{X_1} , f_{X_2} , and f_{X_3} .

$$f_{X_1} = \sum_{x_2} \sum_{x_3} f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \begin{cases} \frac{1}{4} & x_1=0 \\ \frac{3}{4} & x_1=1 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{e.g. when } x_1=0, f_{X_1} = \frac{(0+0+0-1)^2}{8} + \frac{(0+1+0-1)^2}{8} + \frac{(0+0+1-1)^2}{8} + \frac{(0+1+1-1)^2}{8} = \frac{1}{4}.$$

$$f_{X_2} = \begin{cases} \frac{1}{4} & x_2=0 \\ \frac{3}{4} & x_2=1 \\ 0 & \text{otherwise} \end{cases} \quad f_{X_3} = \begin{cases} \frac{1}{4} & x_3=0 \\ \frac{3}{4} & x_3=1 \\ 0 & \text{otherwise} \end{cases}$$

Materials

